

PricePilot AI

Milestone 1 Documentation

Requirements & Dataset Preparation

Title:

PricePilot AI: Dynamic Pricing Optimization & Revenue Intelligence System

Project Objective:

Build an AI-powered dynamic pricing platform that helps businesses optimize product prices based on market demand, competitor pricing, customer behavior, and sales performance. The system should support price prediction, competitor analysis, demand-based pricing, revenue optimization, and pricing analytics through a centralized platform. The platform is designed to maximize revenue, improve profitability, enhance market competitiveness, and support intelligent pricing decisions using machine learning. This solution can be used by e-commerce companies, retail businesses, marketplaces, airlines, hotels, subscription platforms, and sales teams.

Milestone 1 Scope

The project specification defines Week 1–2 as the initialization and core-setup milestone. Activities include defining pricing workflows and objectives, designing the system architecture/database schema, planning UI workflows, setting up environments, loading retail-pricing and e-commerce-sales datasets, and preparing product-management and pricing-dashboard workflows.

Data Source

Datasets Used

Five datasets were reviewed for PricePilot AI. They cover complementary aspects of pricing, products, customer reviews, retail demand, competitor pricing, promotions, inventory, profitability and economic conditions.

Dataset	Rows	Columns	Primary Purpose
Amazon Product Pricing & Reviews	1,465	16	Product price, discount, rating and review information
Amazon Product Pricing	20	11	Product pricing, discount, rating and stock information
Retail Price Optimization	1,040	13	Weekly price, competitor price and sales-volume analysis
Favorita Store Sales	7,300	8	Store/product demand, promotions, holidays and economic indicator
Integrated Pricing & Demand	7,300	31	Consolidated pricing, demand, competition, inventory and profitability dataset

Dataset Integration Rationale

The five datasets were not treated as interchangeable records. Each contributes a different business perspective. The Amazon datasets provide product-level pricing, discount, rating, review and stock context; the

Retail Price Optimization dataset provides competitor-price and weekly-sales relationships; the Favorita dataset provides store-level demand, promotions, holidays and an economic indicator; and the Integrated Pricing & Demand dataset combines the main modelling variables into one structured dataset. The integrated dataset is therefore the most suitable foundation for the subsequent PricePilot AI modelling workflow, while the other datasets are useful for validation, exploratory comparison and feature understanding.

. Data Preprocessing

Step 1: Data Loading

All five CSV datasets were loaded into Pandas DataFrames and inspected for dimensions, column names, data types and sample records.

Step 2: Missing-Value Analysis

Missing-value checks were performed across all five datasets. The currently loaded datasets contain no missing values in their stored records.

Step 3: Duplicate Analysis

Duplicate-row checks were performed for every dataset. No duplicate rows were identified in the current datasets.

Step 4: Data-Type Preparation

Date fields in the Favorita and Integrated Pricing & Demand datasets were converted to datetime format for temporal analysis. Numerical, categorical and identifier fields were separated conceptually for EDA and future modelling.

Dataset-Level Observations

- Amazon Product Pricing & Reviews: 1,465 records and 16 variables. It contains product information, actual/discounted prices, discount percentage, ratings, review counts and review-related fields.
- Amazon Product Pricing: 20 records and 11 variables. It provides product category/sub-category, MSRP, discounted price, discount percentage, cost, rating and stock availability.
- Retail Price Optimization: 1,040 records and 13 variables. It contains unit price, three competitor prices, freight price, product score and weekly sales volume across products and weeks.
- Favorita Store Sales: 7,300 records and 8 variables. It contains date, store, product, product family, sales units, promotion, holiday and economic-index information.
- Integrated Pricing & Demand: 7,300 records and 31 variables. It includes price, cost, discount, promotion, competitor prices, demand, revenue, gross profit, margin, stock, calendar, holiday, economic and sales-channel features.

Exploratory Data Analysis

1 Product Pricing & Discount Analysis

The Amazon product-pricing dataset contains 5 categories and 18 sub-categories. Its fields support comparison of MSRP, discounted price, discount percentage and cost price. The recorded average actual/MSRP price is 105.49, while the average discounted price is 91.01.

The average discount percentage is 14.71%, and the average product rating is 4.60. These variables are useful for studying pricing and perceived product value.

.2 Competitor Pricing & Weekly Sales

The retail-price dataset has 20 products across 5 categories and 52 weekly observations per product. The own price and competitor-price fields are strongly related, making competitor comparison an important feature for pricing decisions.

Average weekly sales volume is 93.48 units. The dataset also includes freight price and product-score variables that can be considered as additional explanatory features.

.3 Store Demand, Promotion & Holiday Analysis

The Favorita dataset contains 1 stores, 20 product IDs and 5 product families. Total recorded sales units are 275,115. Promotion and holiday indicators provide direct contextual variables for demand analysis.

The data spans 2025-01-01 to 2025-12-31, allowing time-based analysis of sales trends and seasonality.

.4 Integrated Pricing, Demand & Revenue Analysis

The integrated dataset contains 20 products across 5 categories. It includes current price, base MSRP, cost price, discount percentage, three competitor prices, competitor averages/minimums, units sold, revenue, gross profit, profit margin, stock level, promotion, holiday, calendar and macro-economic variables.

Across the dataset, total units sold are 293,345 and total recorded revenue is 28,245,978.64. The average current price is 98.94, average discount is 6.20%, and average profit margin is 54.89%.

. Cross-Dataset Business Insights

1. Pricing decisions should consider more than the listed product price. Competitor prices, discounts, demand, stock and product/customer indicators are available across the combined sources.
2. Competitor pricing is a major input for competitive positioning. The Retail Price Optimization and Integrated Pricing & Demand datasets directly support competitor-price comparisons.
3. Promotions and holidays are important demand-context variables. The Favorita and Integrated datasets allow their relationship with sales to be studied.
4. Revenue and profitability can be incorporated into optimization rather than focusing only on sales volume. The Integrated dataset contains revenue, gross profit and profit margin.
5. Inventory information in the Integrated dataset enables future pricing decisions to consider stock availability and stockout conditions.
6. Time-related variables such as day, month, weekend and holiday indicators provide a foundation for demand forecasting.

. Key EDA Findings

Area	Finding / Relevance
Data quality	All five datasets were checked for missing values and duplicate rows; no missing or duplicate records were identified in the current files.
Pricing	Actual/base/current and discounted prices are available across the product-pricing sources.
Competition	Competitor price fields are available in the retail optimization and integrated datasets.
Demand	Weekly sales volume and/or units sold provide demand targets.
Promotion	Promotion/discount fields enable promotion-effectiveness analysis.
Seasonality	Date, week, month, weekend and holiday fields support time-based analysis.
Profitability	The integrated dataset includes revenue, gross profit and profit margin.
Inventory	Stock level and stockout indicators are available in the integrated dataset.
Customer/Product signals	Ratings and review counts are available in the Amazon-oriented datasets.
Economic context	Economic/macro-economic indicators are available in Favorita and the integrated dataset.

Challenges Faced

- **Dataset Integration:** Combining five datasets with different structures and column names.
- **Data Granularity:** Handling product-level, store-level, and time-based data differences.
- **Data Preprocessing:** Cleaning and preparing datasets for analysis.
- **Feature Mapping:** Matching similar features across different datasets.
- **Time Alignment:** Managing different date and weekly time formats.
- **Price–Demand Analysis:** Establishing relationships between pricing and sales data.
- **Data Consistency:** Maintaining accuracy while combining different data sources.
- **Dataset Selection:** Identifying the most suitable dataset for future modelling.
- **EDA Complexity:** Performing exploratory analysis across multiple datasets.
- **Machine Learning Preparation:** Preparing features for future price prediction and demand forecasting.

Outcome of Milestone 1

1. **Project Setup:** Project environment and basic system architecture were established.
2. **Dataset Collection:** Five relevant pricing, sales, and demand datasets were collected.
3. **Data Understanding:** Dataset structures, features, and data types were analyzed.
4. **Data Cleaning:** Missing values, duplicates, and data inconsistencies were checked and handled.
5. **EDA Completed:** Exploratory analysis was performed to identify pricing, sales, promotion, and demand patterns.
6. **Feature Identification:** Key pricing, competitor, sales, promotion, and business features were identified.
7. **Dataset Integration Plan:** A clear approach for using multiple datasets in the PricePilot AI system was established.
8. **Model Preparation:** The cleaned datasets were prepared for future price prediction and demand forecasting.
9. **Dashboard Planning:** Product and pricing dashboard workflows were defined for the next development phase.
10. **Milestone 1 Completion:** The foundation for **PricePilot AI – Dynamic Pricing Optimization & Revenue Intelligence System** was successfully established

Conclusion

Milestone 1 established the data and analytical foundation for PricePilot AI. The five datasets collectively provide complementary information about product pricing, discounts, competitor prices, customer/product signals, store demand, promotions, seasonality, inventory, revenue, profitability and economic conditions. The exploratory analysis confirms that the Integrated Pricing & Demand dataset is the strongest central dataset for the next modelling stage because it combines pricing, competition, demand, profitability, inventory and contextual variables. The remaining datasets provide valuable supporting perspectives for pricing and demand intelligence. This foundation prepares the project for Milestone 2: Price Prediction and Demand Forecasting.

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